

## URYUPINSK, Russian Federation

WMO: 342400

Lat: 50.800N

Lon: 42.000E

Elev: 347

StdP: 14.51

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -9.5       | -4.0 | -15.0                         | 2.5 | -8.5 | -9.1 | 3.4 | -3.4 | 24.2                  | 25.9 | 22.4 | 27.4 | 4.0                      | 330  | 0.571 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.8                   | 92.4            | 68.0 | 88.6 | 67.2 | 85.3 | 65.8 | 71.1                | 86.1 | 69.5 | 83.4 | 68.0 | 81.1 | 8.9                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 98.7 | 75.5 | 64.9 | 93.3 | 74.3 | 63.4 | 88.6 | 73.7 | 35.1          | 86.7 | 33.8 | 83.5 | 32.5 | 81.4 | 80.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 22.4          | 20.3 | 18.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 45.6  | 20.4  | 21.8  | 30.3  | 48.4  | 59.0  | 68.2  | 71.6  | 69.0  | 57.5  | 44.6  | 31.7  | 23.5  | (6)  |
| (7)  |                                                                                  | DBStd       | 20.79 | 10.70 | 11.41 | 8.88  | 8.08  | 8.56  | 6.83  | 6.02  | 6.99  | 8.49  | 8.96  | 11.10 | 12.20 | (7)  |
| (8)  |                                                                                  | HDD50       | 4076  | 917   | 789   | 610   | 121   | 19    | 0     | 0     | 0     | 30    | 218   | 551   | 821   | (8)  |
| (9)  |                                                                                  | HDD65       | 7658  | 1382  | 1209  | 1075  | 497   | 229   | 47    | 9     | 40    | 250   | 635   | 999   | 1286  | (9)  |
| (10) |                                                                                  | CDD50       | 2478  | 0     | 0     | 0     | 74    | 297   | 545   | 668   | 588   | 255   | 49    | 2     | 0     | (10) |
| (11) |                                                                                  | CDD65       | 584   | 0     | 0     | 0     | 0     | 42    | 142   | 212   | 162   | 25    | 1     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74       | 6128  | 0     | 0     | 0     | 2     | 403   | 1383  | 2264  | 1745  | 318   | 13    | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80       | 2290  | 0     | 0     | 0     | 0     | 85    | 469   | 961   | 697   | 77    | 1     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg       | 8.3   | 8.4   | 9.7   | 9.8   | 9.5   | 8.4   | 7.3   | 6.7   | 7.1   | 7.4   | 7.6   | 8.6   | 8.8   | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 18.0  | 1.3   | 1.1   | 1.1   | 1.0   | 1.6   | 2.3   | 2.1   | 1.3   | 1.9   | 1.5   | 1.5   | 1.4   | (15) |
| (16) |                                                                                  | PrecMax     | 25.0  | 2.4   | 2.4   | 2.2   | 2.4   | 3.4   | 4.9   | 5.9   | 3.3   | 4.6   | 3.5   | 3.0   | 2.7   | (16) |
| (17) |                                                                                  | PrecMin     | 12.2  | 0.6   | 0.4   | 0.2   | 0.2   | 0.4   | 0.5   | 0.2   | 0.1   | 0.2   | 0.3   | 0.1   | 0.3   | (17) |
| (18) |                                                                                  | PrecStd     | 3.6   | 0.5   | 0.5   | 0.6   | 0.6   | 0.7   | 1.1   | 1.4   | 0.8   | 1.3   | 0.8   | 0.8   | 0.7   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 42.5  | 42.3  | 54.3  | 73.8  | 85.6  | 92.7  | 97.7  | 95.2  | 86.3  | 74.9  | 58.3  | 47.7  | (19) |
| (20) |                                                                                  |             | MCWB  | 41.8  | 40.1  | 46.0  | 54.9  | 62.4  | 67.0  | 69.7  | 68.9  | 62.7  | 59.5  | 53.4  | 46.5  | (20) |
| (21) |                                                                                  | 2%          | DB    | 37.7  | 38.5  | 48.8  | 69.6  | 82.3  | 88.2  | 93.7  | 91.7  | 82.2  | 67.7  | 53.5  | 42.6  | (21) |
| (22) |                                                                                  |             | MCWB  | 36.2  | 36.4  | 42.6  | 53.7  | 61.3  | 67.9  | 68.8  | 67.3  | 61.1  | 57.0  | 50.7  | 41.0  | (22) |
| (23) |                                                                                  | 5%          | DB    | 35.3  | 36.5  | 44.7  | 66.1  | 78.5  | 85.4  | 89.4  | 87.5  | 77.9  | 61.6  | 49.8  | 39.2  | (23) |
| (24) |                                                                                  |             | MCWB  | 33.9  | 34.7  | 40.3  | 52.3  | 61.1  | 66.8  | 67.8  | 65.8  | 60.3  | 54.0  | 47.3  | 37.5  | (24) |
| (25) |                                                                                  | 10%         | DB    | 33.6  | 34.9  | 41.2  | 62.8  | 74.3  | 82.2  | 85.3  | 83.5  | 72.6  | 57.4  | 45.8  | 36.5  | (25) |
| (26) |                                                                                  |             | MCWB  | 32.3  | 33.2  | 38.2  | 51.3  | 59.6  | 66.0  | 66.6  | 64.7  | 58.6  | 51.2  | 43.8  | 34.9  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 41.7  | 40.7  | 47.7  | 58.4  | 67.4  | 73.8  | 72.6  | 71.9  | 66.5  | 61.5  | 54.5  | 46.9  | (27) |
| (28) |                                                                                  |             | MCD B | 42.6  | 42.0  | 52.4  | 66.5  | 76.2  | 85.0  | 89.8  | 86.3  | 81.5  | 70.6  | 57.9  | 47.5  | (28) |
| (29) |                                                                                  | 2%          | WB    | 36.9  | 37.0  | 43.7  | 56.4  | 64.9  | 70.9  | 71.1  | 70.1  | 64.0  | 58.8  | 50.9  | 41.1  | (29) |
| (30) |                                                                                  |             | MCD B | 37.6  | 38.1  | 47.9  | 64.2  | 75.7  | 81.9  | 88.4  | 84.7  | 74.8  | 66.0  | 53.2  | 42.5  | (30) |
| (31) |                                                                                  | 5%          | WB    | 34.0  | 34.9  | 40.8  | 54.7  | 62.7  | 68.7  | 69.8  | 68.4  | 61.8  | 55.2  | 47.5  | 37.7  | (31) |
| (32) |                                                                                  |             | MCD B | 34.9  | 36.1  | 44.1  | 62.8  | 72.9  | 80.2  | 84.6  | 82.5  | 73.0  | 60.7  | 49.6  | 38.9  | (32) |
| (33) |                                                                                  | 10%         | WB    | 32.6  | 33.5  | 38.5  | 52.7  | 60.6  | 67.2  | 68.2  | 66.6  | 59.9  | 51.7  | 44.1  | 35.0  | (33) |
| (34) |                                                                                  |             | MCD B | 33.7  | 34.7  | 41.1  | 61.1  | 72.0  | 78.6  | 82.3  | 79.2  | 70.6  | 56.6  | 46.0  | 36.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 9.6   | 10.7  | 11.8  | 17.6  | 18.2  | 19.1  | 20.8  | 22.2  | 19.6  | 13.6  | 9.5   | 8.6   | (35) |
| (36) |                                                                                  |             | MCDBR | 7.1   | 8.5   | 16.7  | 23.7  | 24.0  | 24.8  | 26.8  | 27.2  | 28.3  | 20.8  | 11.1  | 6.9   | (36) |
| (37) |                                                                                  |             | MCWBR | 6.4   | 7.5   | 11.1  | 12.1  | 9.7   | 9.5   | 8.8   | 9.0   | 12.6  | 11.4  | 9.4   | 6.5   | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 6.7   | 8.1   | 15.3  | 19.3  | 19.3  | 21.0  | 24.0  | 23.1  | 23.0  | 19.2  | 11.4  | 6.8   | (38) |
| (39) |                                                                                  |             | MCWBR | 6.4   | 7.6   | 11.2  | 11.1  | 9.0   | 9.2   | 9.1   | 9.1   | 12.0  | 11.3  | 10.0  | 6.6   | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.313 | 0.349 | 0.375 | 0.441 | 0.407 | 0.397 | 0.423 | 0.437 | 0.391 | 0.364 | 0.341 | 0.315 | (40)  |      |
| (41) |                                                                                  | taud        | 2.103 | 2.001 | 2.077 | 2.083 | 2.282 | 2.344 | 2.272 | 2.214 | 2.319 | 2.336 | 2.263 | 2.158 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 213   | 235   | 256   | 252   | 269   | 273   | 263   | 252   | 251   | 234   | 205   | 193   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 25    | 37    | 42    | 47    | 40    | 38    | 40    | 41    | 33    | 27    | 22    | 21    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 306   | 631   | 1060  | 1385  | 1838  | 1956  | 1919  | 1686  | 1150  | 610   | 268   | 177   | (44)  |      |
| (45) |                                                                                  | RadStd      | 69    | 116   | 167   | 125   | 167   | 168   | 144   | 128   | 134   | 91    | 55    | 43    | (45)  |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (5 neighbors) | +1.06 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -247  | N/A   | N/A   |

Nomenclature: See separate page